

CUGG1004

Data Preprocessing and EDA

Module-wise Syllabus Breakdown

5 Modules • Theory, Hands-on Practice & Relevance for Each

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MODULE 1

THEORY

Preprocessing Basics



Data Formats & Cleaning

Ingesting raw CSV/JSON sources and resolving duplicates, inconsistent types, and malformed records before any analysis begins.



Missing & Categorical Data

Imputing gaps with mean, median, or mode, then converting categorical labels into numeric form via one-hot and label encoding.



Outliers & Scaling

Detecting anomalies using Z-score and IQR, then normalizing or standardizing so no single feature dominates by scale alone.

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MODULE 1

Preprocessing Basics

PRACTICE

1.1

Load CSV/JSON & identify missing values

Builds the first habit: profiling a dataset's shape and gaps before touching it.

1.2

Imputation: mean, median, mode

Compares strategies for filling gaps without distorting the original distribution.

1.3

One-hot & label encoding

Translates categorical text into numeric inputs that models can consume.

1.4

Outlier detection: Z-score & IQR

Flags extreme values that could skew statistics or destabilize model training.

1.5

Min-Max normalization & standardization

Rescales features onto comparable ranges for distance-based algorithms.

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MODULE 1

Preprocessing Basics

RELEVANCE



Practitioners commonly cite data cleaning as the single largest time-cost in a data science project — making this the highest-leverage skill in the course.



"Garbage in, garbage out": even advanced models fail on dirty inputs, so clean data is a prerequisite, not an afterthought.



Every later module assumes clean, scaled data — feature engineering, visualization, and modeling all build on this foundation.



Establishes reproducible cleaning habits that scale directly to the real-world datasets used in Module 4.



Module 1

*is the foundation every other
module depends on*

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MODULE 2

Feature Engineering

THEORY



Feature Selection

Using correlation analysis and statistical tests to find which variables carry real predictive signal versus noise.



Dimensionality Reduction

Applying PCA and t-SNE to compress high-dimensional data into fewer components while preserving structure.



Feature Creation

Engineering interaction and polynomial terms, then ranking importance to surface relationships models can't find alone.

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MODULE 2

Feature Engineering

PRACTICE

2.1

Feature selection via correlation analysis

Reveals which features move together and which are redundant.

2.2

Statistical feature selection methods

Applies hypothesis tests to rank predictive power objectively.

2.3

Principal Component Analysis (PCA)

Reduces dimensionality while retaining maximum variance for downstream models.

2.4

Dimensionality reduction with t-SNE

Projects complex data into 2D/3D space for visual cluster discovery.

2.5

Polynomial features & interaction terms

Captures non-linear relationships a plain linear model would miss.

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MODULE 2

Feature Engineering

RELEVANCE

- ✓ Well-engineered features routinely beat algorithm tuning — domain knowledge encoded as a feature often outperforms a fancier model on raw data.
- ✓ Dimensionality reduction directly counters the curse of dimensionality, where excess features cause overfitting and slow training.
- ✓ Leaner feature sets mean faster training and lighter models — a real advantage when deploying at scale.
- ✓ PCA and t-SNE outputs feed straight into Module 3's visual storytelling, linking the two modules directly.



**Fewer, smarter
features**

*often beat more complex
models*

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MODULE 3

Data Visualization

THEORY



Visualization Tools

Matplotlib and Seaborn form the core Python plotting stack used across research and production analytics.



Plot Types

Matching the right chart — bar, line, scatter, histogram, heatmap — to the question being asked, not just the data on hand.



Interactive & Insights

Building Plotly dashboards and correlation heatmaps so stakeholders can explore data and act on it directly.

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MODULE 3

Data Visualization

PRACTICE

3.1

Matplotlib: line, bar, scatter plots

Builds fluency with the foundational plotting API most visualization tools wrap.

3.2

Seaborn: histograms, box plots

Adds statistical plotting with cleaner defaults and built-in aggregation.

3.3

Heatmaps for correlation analysis

Surfaces hidden multivariate relationships at a single glance.

3.4

Interactive visualizations with Plotly

Adds zoom, hover, and filter interactivity for exploratory dashboards.

3.5

Decision-driven visualization

Connects chart choice back to the actual business question being answered.

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MODULE 3

Data Visualization

RELEVANCE



A model's findings are only as useful as a stakeholder's ability to see and trust them — visualization bridges analysis and action.



Visual EDA catches issues like skew, leakage, and drift that summary statistics alone can hide.



Interactive dashboards turn a one-time analysis into a reusable decision-making tool for non-technical teams.



Visual communication is consistently ranked among the most in-demand applied data science skills by employers.



Charts

*turn raw numbers into
decisions people act on*

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MODULE 4

Case Studies & Tools

THEORY



Pipelines & AutoML

Scikit-learn Pipelines chain preprocessing and modeling into one reproducible object; AutoML automates model search.



Real-World Data Analysis

Working with messy, undocumented open datasets (Kaggle/UCI) instead of clean teaching data closes the gap to real practice.



Group Projects

Collaborative EDA work mirrors how data teams actually divide, review, and present exploratory analysis.

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MODULE 4

Case Studies & Tools

PRACTICE

4.1

Build preprocessing pipelines with Scikit-learn

Chains cleaning, encoding, and scaling into one repeatable, leak-proof object.

4.2

Data cleaning on a real-world dataset

Applies Module 1 skills to data that has no answer key.

4.3

Analyze open datasets (Kaggle/UCI)

Builds the habit of independently sourcing and vetting external data.

4.4

Demonstrate basic AutoML tools

Benchmarks automated model search against manual feature engineering.

4.5

Mini group EDA project

Simulates a real analytics sprint with peer review and presentation.

Case Studies & Tools

RELEVANCE

- ✓ Pipelines prevent a common production bug: data leakage caused by inconsistent preprocessing between train and test sets.
- ✓ AutoML literacy helps analysts judge when to automate and when domain expertise still wins.
- ✓ Open-dataset work produces a portfolio piece — directly useful for interviews and resumes.
- ✓ Group EDA projects build the communication and division-of-labor skills that data roles require every day.



**Theory →
Practice**

*this module is the course's
reality check*

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MODULE 5

THEORY

Advanced Data Handling



Time & Text Data

Resampling and smoothing time series, then cleaning text through tokenization, stop-word removal, stemming, and lemmatization.



Imbalanced & Augmentation

Correcting skewed class distributions with SMOTE and ADASYN, and expanding limited data through augmentation.



Automation at Scale

Using Pandas and Dask to automate preprocessing pipelines that scale beyond a single machine's memory.

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MODULE 5

Advanced Data Handling

PRACTICE

5.1

Time-series preprocessing: resampling, smoothing

Handles irregular sampling and noise before forecasting or trend analysis.

5.2

Text preprocessing: tokenization, stemming, lemmatization

Converts raw language into clean tokens models can actually use.

5.3

Handle imbalanced data: SMOTE & ADASYN

Balances rare-class data so models stop ignoring minority outcomes.

5.4

Data augmentation for text/image data

Synthesizes variation when collecting more real data isn't possible.

5.5

Automated pipelines with Pandas & Dask

Scales preprocessing from a laptop to large, distributed datasets.

Advanced Data Handling



Most real-world data is messy, unstructured, or imbalanced — clean, balanced datasets are the classroom exception, not the rule.



Fraud detection, medical diagnosis, and churn prediction all depend on correctly handling imbalanced classes.



Text and time-series skills extend the course directly into NLP and forecasting applications.



Dask and automation skills prepare students for big-data settings where Pandas alone won't fit in memory.



**Real data is
messy**

*this module makes that
survivable*